

Car Rental Fleet Manager - Module 3

Only Module 3 parts: methods, arrays, linear search and recursion.

```
// MODULE 3: Methods, Recursion & Arrays
// This part organizes the project using methods and arrays.
// It also adds searching and a simple recursion example.
```

```
// Store all cars in an array
```

```
Car[] cars = {
    new Car(101, "GTR", 5000),
    new Car(102, "Porsche", 4500),
    new Car(103, "Bugatti", 3000)
};
```

```
// Method to display all cars
```

```
static void displayCars(Car[] cars) {
    for (Car car : cars) {
        car.display();
    }
}
```

```
// Method to find a car using its ID
```

```
static Car findCar(Car[] cars, int id) {
    for (Car car : cars) {
        if (car.id == id) {
            return car;
        }
    }
    return null;
}
```

```
// Recursive method to count available cars
```

```
static int countAvailable(Car[] cars, int index) {

    if (index == cars.length) {
        return 0;
    }

    int count = 0;

    if (cars[index].available) {
        count = 1;
    }

    return count + countAvailable(cars, index + 1);
}
```